



Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin

Ollscoil Átha Cliath | The University of Dublin

FACULTY OF ARTS, HUMANITIES AND SOCIAL SCIENCES

SCHOOL OF SOCIAL SCIENCES AND PHILOSOPHY

DEPARTMENT OF ECONOMICS

Senior Fresh
BESS, TJH, PPES

Hilary Term 2021

EC: ECONOMICS 3

FOUNDATION SCHOLARSHIP EXAMINATION

Friday 29th January 2021 14.00 – 16.15

Professor Alan Matthews, Professor Tara Mitchell, Professor Davide Romelli

Instructions to Candidates:

Please answer ANY three questions. Each question carries equal weight.

Maximum word limits (of 1,500 words for each of the answers) must be obeyed; readers will only read (and hence grade) up to the maximum word count.

Materials permitted for this examination:

None required.

You may not start this examination until you are instructed to do so by the Invigilator.



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Foundation Scholarship Examinations 2020/21

Cover sheet

Please complete this cover sheet and include it with your exam paper submission/answers for upload in a single file.

Student Exam Number:	
Paper Title:	
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Enter an "X" in the boxes below to indicate that you understand the regulations with regard to plagiarism and that you have completed the appropriate training.

I have read and I understand the plagiarism provisions in the General Regulations of the University Calendar for the current year, found at <http://www.tcd.ie/calendar>.

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Exam questions begin on page 3 below.

Section A**1.**

- a) A pharmaceutical company has developed a new drug and has a patent which grants it monopoly rights to produce it. Represent this situation graphically showing the isoprofit curves and marginal cost curve for the firm and the market demand curve. Use the diagram to demonstrate what the quantity and price will be in this market in equilibrium and explain how this will be chosen.

[30 marks]

- b) After 10 years, the patent expires, and other firms are allowed to enter the market and produce generic versions of the drug. Using graphs, demonstrate what this new situation will look like both for an individual firm in this market and the market overall, assuming that the market for this drug is now perfectly competitive. Explain what will happen to the equilibrium price and quantity in this market after the patent expires and why.

[30 marks]

- c) Discuss the reasons why, in practice, the market for this drug may not be perfectly competitive even after the initial patent expires.

[40 marks]

2. Two major supermarkets are trying to decide what price to charge for their Christmas turkeys. Suppose each can charge either a high price or a low price. If both supermarkets choose a low price, they will both earn low profits. If both supermarkets choose a high price, they will both earn medium profits. If one supermarket chooses a low price and the other chooses a high price, then the supermarket that chose the low price will earn high profits and the supermarket that chose the high price will earn very low profits (high profits > medium profits > low profits > very low profits). Assume that the supermarkets only care about their own profits.

- a) Draw the decision box for this game and find the Nash equilibrium. Explain why there is a unique equilibrium in this game.

[30 marks]

- b) Suppose that the relevant directors of the two supermarkets can talk to each other before making the pricing decision and can make a verbal agreement to choose a high price. However, they cannot write a binding contract and must choose their actual prices simultaneously. Explain how this might change the outcome of the game? What factors might influence whether or not the firms are able to successfully sustain collusive behaviour? Who would lose out in this case if the firms were successful?

[35 marks]

- c) Suppose now that customers will go to the supermarket that has the lowest price turkey and, in addition to buying the turkey, they will do all of their Christmas grocery shopping in that supermarket. How might this change the payoffs to the supermarkets? Would this make it more or less likely that they would successfully be able to collude? Explain your answer.

[35 marks]

3.

- a) An individual has a job where she gets paid 20 euro per hour and has to choose how many hours to work. The individual likes leisure time but also likes the consumption that she gets from the income that she earns. Use a diagram to represent the indifference curves and budget constraint of the individual and show how she will choose the number of hours to work. Explain your answer.

[30 marks]

- b) Suppose the individual gets a raise and now earns 40 euro per hour. Will she now work more hours or fewer hours than before? Use a diagram and the concepts of income and substitution effects to explain your answer.

[35 marks]

- c) Do you think that this model of consumer choice is generally useful for understanding how individuals make choices about how many hours to work? Would it be a more or less useful model for explaining why average working hours are different in different countries? Explain your answers.

[35 marks]

Section B

4.

a) The imminent approval of a Covid-19 vaccine is causing a boost in business confidence among Irish firms, such that investment demand increases for any interest rate. Use the AD-AS diagram to explain how this situation will affect the price level and real GDP in Ireland in the short-run. (Assume the economy is initially at its long run equilibrium level of output.) What happens to the unemployment rate? Use sticky wage theory of aggregate supply to explain what will happen to output and the price level in the long run (assuming there is no change in policy). [45 marks]

b) Germany has run a trade surplus over the past decades. Has this surplus been good or bad for the German economy? Be sure to discuss the implied change in net capital outflow and the advantages/disadvantages of a trade surplus. [55 marks]

5.

a) In response to the failed negotiations for a soft Brexit agreement, suppose the Bank of England raises the rate of money growth permanently. Using models developed in class, explain clearly the short-run and long run effects of such a policy on i) real GDP ii) the price level iii) inflation and iv) the real exchange rate. Explain clearly whether you would recommend this policy. [55 marks]

b) Suppose an economy is in long-run equilibrium. If the domestic nominal exchange appreciates, use the AS-AD model to examine the short and long-run implications for i) output and ii) the real exchange rate. [45 marks]

6.

- a) Explain briefly why both the money multiplier and the government spending multiplier might be lower in a recession. What are the implications of this for economic policy? [25 marks]
- b) According to the monetary approach to the exchange rate, a permanent rise in the money supply would have no effect on the real exchange rate. Explain how this implication would arise. [35 marks]
- c) Assume the economy is in long-run equilibrium, and everyone saves half of their income. Use the AS-AD model to show how, in the short run, an increase in investment demand can cause an increase in savings. Was this a feature of the loanable funds model? Explain your answer. [40 marks]